

Chromosomal Disorders

Extracted & reformatted from: MedStar Pediatrics, 2nd Edition — Chapter 13: Genetic Disorders and Dysmorphic Features, Section 13.1 (pp. 566–582). All original content preserved verbatim; only layout/typography changed for readability.

DYSMORPHIC FEATURE

- A dysmorphic feature is a difference of body structure.
 - It can be an isolated finding in an otherwise normal individual, or it can be related to a congenital disorder, genetic syndrome, or birth defect.
 - One of the key challenges in identifying and describing dysmorphic features is the use and understanding of specific terms between different individuals.
 - Clinical geneticists and pediatricians are usually those most closely involved with the identification and description of dysmorphic features, as most are apparent during childhood.
 - Dysmorphic features are invariably present from birth, although some are not immediately apparent upon visual inspection.
- Common Chromosomal Disorders associated with Dysmorphic Features
 - Down Syndrome
 - Edward Syndrome
 - Turner Syndrome

DOWN SYNDROME

- Trisomy 21 is the most common and best-known chromosomal disorder in humans and the most common cause of intellectual disability.

Epidemiology and Risk Factor

- Incidence: (1:733); prevalence are increasing b/c of increase in life span in the last decade.
- Age-related demographics; occurrence is strongly dependent on maternal age.

Maternal Age	Risk
15–29 years	1 case in 1500 live births
30–34 years	1 case in 800 live births
35–39 years	1 case in 270 live births
40–44 years	1 case in 100 live births
>45 years	1 case in 50 live births

- Sex-related demographics; M>F (1.15:1)
NB: 50% of female patients with trisomy 21 are fertile and these females have up to 50% chance of having a live child who also has trisomy 21. On the other hand, men with down syndrome are usually infertile, except for those with mosaicism.
- Race-related demographics; found in all race (African American patients live shorter life spans than white patients)

Types of Down Syndrome

Down syndrome is caused by the following 3 cytogenic variants:

Variant	Prevalence	Key notes
1. Three full copies of chromosome 21 (trisomy 21)	95%	Origin: maternal non-disjunction, with meiosis I in 97% and the rest are paternal.
2. Chromosomal translocation resulting in 3 copies of the DS critical region	4%	Carries high recurrence rate relative to other forms — 100% for translocation 21;21; other t(15;21) have
3. Mosaicism (some cells having 46 chromosomes)	1%	Patients who are mosaic tend to have a milder phenotype.

Clinical Manifestation

History

When recording the history from the parents of a child with Down syndrome, the clinician should include the following:

- Parental concern about hearing, vision, developmental delay, respiratory infections, and other problems.
- Vomiting — secondary to GI tract blockage by **duodenal web or atresia**
- Absence of stools secondary to **Hirschsprung disease**
- Diaphoresis while **feeding**, **arrhythmia**, fainting **episodes**, palpitations, or chest pain secondary to heart lesion
- Symptoms of sleep apnea, including snoring, restlessness during sleep, difficulty awaking, daytime somnolence, behavioral changes, and school problems (because **of macroglossia**)
- Symptoms of **atlantoaxial instability** — may cause vague neck pain, occipital headache, or occasionally intermittent (and potentially fatal) cervical spinal cord compression, including:
 - Easy fatigability, limited neck mobility or head tilt, torticollis, difficulty walking, change in gait pattern, loss of motor skills, incoordination, clumsiness, sensory deficits, spasticity
 - Hyperreflexia, clonus, extensor-plantar reflex, loss of upper-body strength, abnormal neurologic reflexes, change in bowel and bladder function, increased muscle tone in the legs, and changes in sensation in the hands and feet
- Delay in cognitive abilities, motor development, language development (specifically expressive skills), and social competence
- Assess feeding history to ensure adequate caloric intake
- Prenatal diagnosis of Down syndrome
- Maternal and paternal age (risk factor)
- Family history of similar illness b/c of recurrence
- In rare cases, the symptoms progress to paraplegia, hemiplegia, quadriplegia.

Physical Examination — Dysmorphic Features

- **Upslanting palpebral fissures**, epicanthic folds, and brachycephaly are nearly universal features of DS.
- The other dysmorphic features of DS are each present in 47 to 82% of cases.
- These features predominantly affect the head and neck and the extremities.



Fig. 103 — Brushfield spots, visible between the inner and outer circles of the iris.



Fig. 104 — Facial features of Down syndrome.

Head and Neck

- Brushfield spots
- Upslanting palpebral fissures
- Epicanthic folds
- Flat facial profile / flat nasal bridge
- Folded or dysplastic ears
- Low-set small ears

- Brachycephaly
- Open mouth
- Protruding tongue
- Furrowed tongue
- Short neck
- Excessive skin at nape of the neck
- Narrow palate
- Abnormal teeth

Extremities



Fig. 105 — Sandal gap.



Fig. 106 — Single transverse palmar crease.

- Short broad hands
- Incurved fifth finger with hypoplastic mid phalanx
- Transverse palmar crease
- Space between the first and second toes (Sandal Gap)
- Hyperflexibility of joints

Cardiovascular System

- Endocardial cushion defect
- Ventricular septal defect
- Atrial septal defect
- Patent ductus arteriosus
- Aberrant subclavian artery
- Pulmonary hypertension

Skin



Fig. 107 — Cutis marmorata.

- Cutis marmorata

Neonatal Features

- Ten of the characteristic dysmorphic features are common in newborns with DS and are usually recognized soon after birth.
- These 10 criteria are called **HILL'S CRITERIA**, used to diagnose Down syndrome in the absence of prenatal diagnosis.
- In the presence of 8 criteria we can diagnose a full-blown Down syndrome:

Flat facial profile	Transverse palmar (Simian) crease
Slanted palpebral fissures	Excessive skin at nape of the neck
Small, dysplastic ears	Hyperflexibility of joints
Hypotonia	Dysplasia of pelvis
Poor Moro reflex	Dysplasia of mid-phalanx of fifth finger

Complications

Complications of Down syndrome involve almost every organ system of the body.

HEENT

- Congenital or acquired hearing loss
- Serous otitis media
- Refractive errors (myopia)
- Congenital or acquired cataracts
- Nystagmus
- Strabismus
- Glaucoma
- Blocked tear duct
- Sinusitis and nasopharyngitis may occur secondary to narrow nasal passages and sinuses.

Cardiopulmonary

- Cardiovascular complications are important in Down syndrome:
 - Acquired mitral, tricuspid, or aortic valve regurgitation
 - Endocarditis
 - Pulmonary arterial hypertension
 - Mitral valve prolapse
 - Obstructive sleep apnea, secondary to enlarged tonsils or to other causes of upper airway obstruction.

Gastrointestinal

- Celiac disease
- Gastroesophageal reflux
- Infants with oral-motor difficulties may present with choking and gagging on feedings
- Dysphagia may affect children as well as adults
- Chronic constipation
- Delayed tooth eruption

Musculoskeletal

- Atlantoaxial instability — defined as increased mobility of the cervical spine at the level of the first and second vertebrae, can lead to subluxation of the cervical spine
- Hip dysplasia
- Slipped capital femoral epiphyses
- Avascular hip necrosis
- Recurrent joint dislocations (shoulder, knee, elbow, thumb)

Endocrine

- Congenital or acquired hypothyroidism
- Diabetes mellitus
- Infertility
- Obesity
- Hyperthyroidism

Hematologic

- Transient myeloproliferative syndrome
- Acute lymphocytic leukemia
- Acute myelogenous leukemia

Cutaneous

- Hyperkeratosis
- Seborrhea
- Xerosis
- Perigenital folliculitis

Neuropsychiatric

- Developmental delay
- Seizures
- Autism spectrum disorders
- Behavioral disorders (disruptive)
- Depression
- Alzheimer disease (present in almost all individuals with Down syndrome by age 40 years)
- Abuse — individuals with Down syndrome are at high risk for physical and sexual abuse

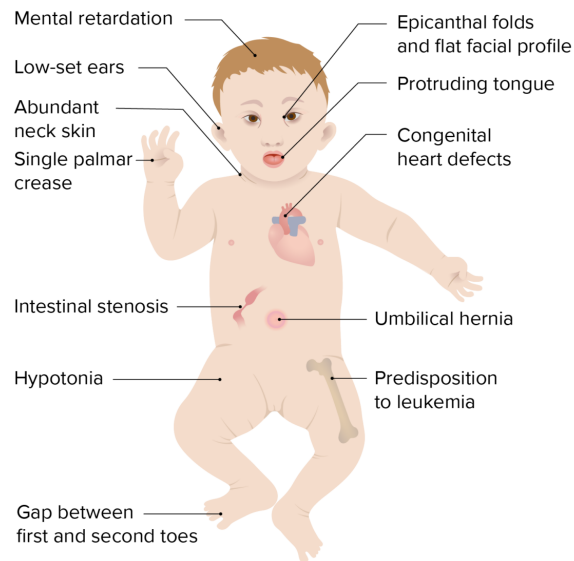


Fig. 108 — Summary of features and complications of Down Syndrome.

Investigations

- The diagnosis of Down syndrome is most commonly made by prenatal screening followed by definitive diagnostic testing.
- When prenatal diagnosis has not been made, Down syndrome is usually apparent from the clinical examination of the newborn.
- Diagnosis should be confirmed through chromosomal analysis.
- Since Down syndrome is associated with multisystem involvement, additional diagnostic studies are performed as appropriate.

Prenatal Screening and Diagnosis

- Prenatal screening using a combination of maternal serum biomarkers and ultrasonography can detect up to 95% of pregnancies affected by Down syndrome. The false positive rate is 5%.
- Recently updated guidelines from the American College of Obstetricians and Gynaecologists (ACOG) state the following:
 - All women should be offered screening for aneuploidy before 20 weeks' gestation
 - All pregnant women, regardless of their age, should have the option of diagnostic testing.

First-Trimester Screening

- For pregnant women, for whom an early diagnosis is important, a first-trimester "combined test" performed at 11–14 weeks involving:
 - Sonographic testing for NT (nuchal translucency), together with
 - Testing for PAPP-A and hCG
- Provides a detection rate of 82–87% for Down syndrome.

Second-Trimester Screening

- Tests used for second-trimester screening include the triple and quadruple screens.
- The triple screen calculates the risk of Down syndrome by measuring:
 - Serum hCG,
 - AFP, and
 - Unconjugated estriol
- Can detect up to 69% of Down syndrome pregnancies.
- Currently, the quadruple test, usually performed at 15–18 weeks' gestation, is the most common screening test performed in the second trimester.
 - This screen measures inhibin A in addition to the biochemical markers measured in the triple screen.

- Provides an 81% detection rate for Down syndrome.
- In addition, the quadruple test serves as a screening test for open neural tube defects (since it involves measurement of AFP) and can also detect trisomy 18.

Integrated Screening

- With integrated screening, the pregnant woman undergoes a first-trimester screening (involving NT testing, PAPP-A, hCG) followed by the quadruple screen in the second trimester.
- This combined screening approach increases the detection rate of Down syndrome to 95%, with a false positive rate of only 5%.

Ultrasonography

- Prenatal ultrasonography may reveal the following in a fetus with Down syndrome, in the second trimester:
 - Absent or hypoplastic nasal bone,
 - Thickened nuchal fold,
 - Echogenic bowel,
 - Shortened long bones, and
 - Pyelectasis

Postnatal Workup

Commonly performed studies in individuals with Down syndrome include the following.

- **Cytogenetic studies** — the clinical diagnosis of trisomy 21 should be confirmed with cytogenetic studies. Karyotyping is essential to determine the risk of recurrence. In translocation Down syndrome, karyotyping of the parents and other relatives is required for proper genetic counselling.
- **Interphase fluorescence in situ hybridization — FISH** may be used for rapid diagnosis of trisomy 21. It can be successful in both prenatal diagnosis and diagnosis in the neonatal period.
- **Measurement of immunoglobulin G** — focuses on identifying deficiencies of subclasses 2 and 4.
- **Radiography and Ultrasonography** —
 - Current evidence does not support performing routine screening radiographs for assessment of potential atlantoaxial instability in asymptomatic children.
 - Cervical radiography (with lateral flexion and extension views) is required to measure the atlantodens distance and to rule out atlantoaxial instability at the age of 3 years.
- MRI is also recommended regularly for evaluation.
- Echocardiography should be performed on all infants suspected of having trisomy 21 to identify congenital heart disease, regardless of findings on physical examination.

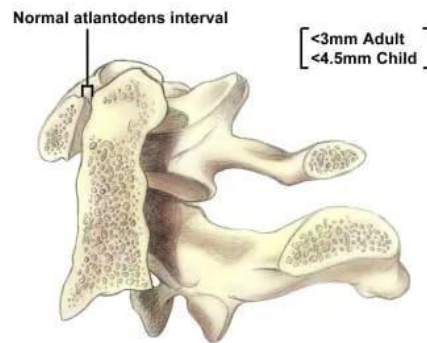


Fig. 109 — Atlantodens instability: normal atlantodens interval (<3 mm adult, <4.5 mm child).

Invasive Diagnostic Tests

- Amniocentesis, routinely performed at 14–16 weeks' gestation, remains the criterion standard of invasive diagnostic tests. Testing for chromosomal disorders is 99.5% accurate.
- Chorionic villus sampling (CVS) is performed at 10–13 weeks' gestation. The accuracy of CVS (96–98%) is less than that of midtrimester amniocentesis, because of confined placental mosaicism and maternal-cell contamination.
- Percutaneous umbilical blood sampling (PUBS) is approximately 95% successful in obtaining a blood sample for cytogenetic testing.

Other Lab Studies

- A complete blood count (CBC) with differential and bone marrow examination to rule out leukaemia is indicated.
- Thyroid-stimulating hormone (TSH) and thyroxine (T4) levels should be obtained at birth, at 6 and 12 months, and annually thereafter, to rule out hypothyroidism.

Principle of Treatment

- Management of Down syndrome requires an organized approach to ongoing evaluation and monitoring for associated abnormalities and prevention of common disorder.

EDWARD SYNDROME (TRISOMY 18)

- Among live born children, trisomy 18 is the second most common autosomal trisomy after trisomy 21.
- The disorder is characterized by: severe psychomotor and growth retardation, microcephaly, microphthalmia, malformed ears, micrognathia or retrognathia, microstomia, distinctively clenched fingers, and other congenital malformations.
- Trisomy 18 severely affects all organ systems.
- In translocations that result in partial trisomy or in cases of mosaic trisomy 18, clinical expression is less severe, and survival is usually longer.

Epidemiology and Risk Factor

- Prevalence: 1 in 6000–8000 live births

Mortality / Morbidity

Approximately 95% of conceptuses with trisomy 18 die as embryos or fetuses; 5–10% of affected children survive beyond the first year of life.

- Race — Trisomy 18 has no racial predilection.
- Sex — Approximately 80% of trisomy 18 cases occur in females.
- Age — Trisomy 18 is detectable during the prenatal and newborn periods.

Risk Factors

- The incidence rate increases with advanced maternal age.
- In approximately 90% of cases, the extra chromosome is maternal in origin, with meiosis II errors occurring twice as frequently as meiosis I errors.
- This is in contrast to other human trisomies, which exhibit a higher frequency of nondisjunction in maternal meiosis I.

Type of Trisomy 18

- Full trisomy 18 is responsible for 95% of Edwards syndrome cases.
- Mosaicism and translocations cause few cases.

NB: An extra chromosome 18 is responsible for the phenotype.

Common Physical Finding



Fig. 110 — Infant with trisomy 18 and overlapping fingers.



Fig. 111 — Rocker-bottom feet.

- Small and premature appearance
- Tight palpebral fissures
- Narrow nose and hypoplastic nasal alae
- Narrow bifrontal diameter
- Prominent occiput
- Micrognathia
- Cleft lip or palate
- Microcephaly
- Congenital heart disease (e.g., VSD, PDA, ASD)
- Short sternum, small nipples
- Limited hip abduction
- Clinodactyly and overlapping fingers; index over 3rd, 5th over 4th; closed fist

- Rocker-bottom feet and hypoplastic nails
- Severe developmental delays and prenatal and postnatal growth restriction
- Premature birth, polyhydramnios
- Inguinal or abdominal hernias
- **NB:** Only 5% live >1 year

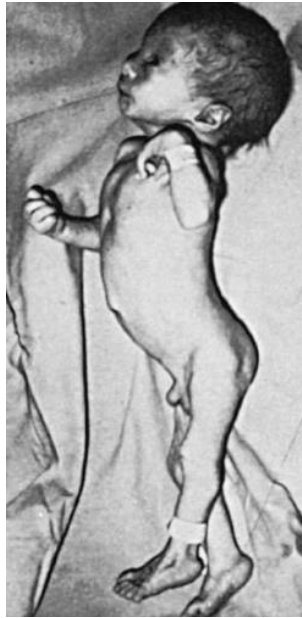


Fig. 112 — Male infant with Trisomy 18 at age 4 days. Note the prominent occiput, micrognathia, low-set ears, short sternum, narrow pelvis, prominent calcaneus, and flexion abnormalities of the finger.

Workup

- First trimester non-invasive screening based on maternal age, serum markers, and sonographic "soft markers" have demonstrated a high sensitivity for the diagnosis of trisomy 18.
- Low levels of human chorionic gonadotrophin (hCG) and low unconjugated estriol (uE3) in maternal serum during mid trimester are useful predictors for an increased risk for trisomy 18.

Prenatal Diagnosis

- Amniocentesis
- Chorionic villus sampling (CVS)
- Percutaneous umbilical blood sampling (PUBS)

Postnatal Diagnosis

- Hematological studies in patients with trisomy 18 during the first week of life:
 - Thrombocytopenia
 - Neutropenia
 - Abnormal erythrocyte values — anemia was detected in 40%, and polycythemia was detected in 17%.
- Conventional Cytogenetic and Fluorescence In Situ Hybridization (FISH) studies —
 - FISH for rapid diagnosis (most laboratories, ≤ 24 hours) is more sensitive for mosaicism in the neonatal period (if unknown prenatally), followed by karyotyping, which is necessary even if FISH confirms the diagnosis for the rare translocation;
 - Karyotyping is also necessary if the diagnosis is made prenatally to confirm the type of trisomy 18.

Principle of Treatment

- Management of Edward syndrome requires an organized approach to ongoing evaluation and monitoring for associated abnormalities and prevention of common disorder.

TURNER SYNDROME (45, X)

- Turner syndrome is a condition characterized by complete or partial monosomy of the X chromosome and defined by a combination of phenotypic features.
- Half of the patients with Turner syndrome have a 45, X chromosome complement.
- The other half exhibits mosaicism and varied structural abnormalities of the X or Y chromosome.

Epidemiology and Risk Factor

- Prevalence: 1 in 5,000 female live births
 - Sex: occurs only in females
 - Race: no racial or ethnic predilections are known.
- NB:** Maternal age is not a predisposing factor for children with 45, X.
- In 75% of patients, the lost sex chromosome is of paternal origin (whether an X or a Y).
 - It has been estimated that 95–99% of 45, X conceptions are miscarried.

Common Finding Associated with Turner Syndrome

- Clinical findings in the newborns can include: small size for gestational age, webbing of the neck, protruding ears, lymphedema of the hands and feet.
- Older children and adults have short stature and exhibit variable dysmorphic features.
 - Congenital heart defects (40%) and structural renal anomalies (60%) are common.
 - The most common heart defects are bicuspid aortic valves, aortic stenosis, and mitral valve prolapse.
 - The gonads are generally streaks of fibrous tissue (gonadal dysgenesis).
 - There is primary amenorrhea and lack of secondary sex characters.
 - These children should receive regular endocrinologic testing.
 - Most patients tend to be of normal intelligence, but intellectual disability is seen in up to 6% of affected children.
 - They are also at increased risk for behavioral problems and deficiencies in spatial and motor perception.
 - Patients with 45,X/46,XY mosaicism can have Turner syndrome, although this form of mosaicism can also be associated with male pseudo-hermaphroditism, male or female genitalia in association with mixed gonadal dysgenesis, or a normal male phenotype.
 - This variant is estimated to represent approximately 6% of patients with mosaic Turner syndrome.

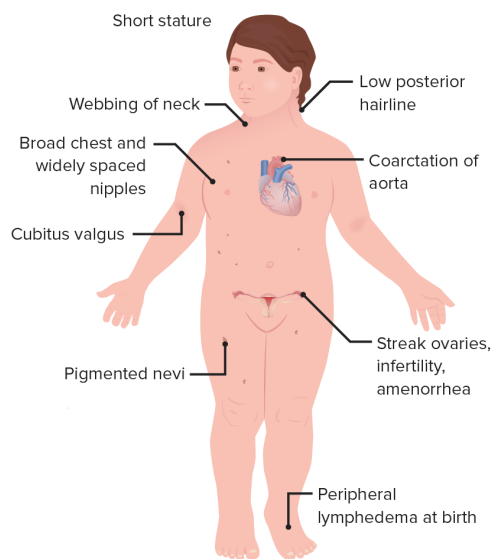


Fig. 113 — Features of Turner syndrome.

Signs Associated with Turner Syndrome

- | | |
|--|---|
| • Short stature | • Gonadal dysgenesis (infertility, primary amenorrhea) |
| • Congenital lymphedema | • Gonadoblastoma (increased risk if Y chromosome material is present) |
| • Horseshoe kidneys | • Learning disabilities (nonverbal perceptual motor and visuospatial skills) (in 70%) |
| • Patella dislocation | • Developmental delay (in 10%) |
| • Increased carrying angle of elbow (cubitus valgus) | • Social awkwardness |
| • Madelung deformity (chondrodysplasia of distal radial epiphysis) | • Hypothyroidism (acquired in 15–30%) |
| • Congenital hip dislocation | • Type 2 diabetes mellitus (insulin resistance) |
| • Scoliosis | • Strabismus |
| • Widespread nipples | • Cataracts |
| • Shield chest | • Red-green color blindness (as in males) |
| • Redundant nuchal skin (in utero cystic hygroma) | • Recurrent otitis media |

- Low posterior hairline
- Coarctation of aorta
- Bicuspid aortic valve
- Cardiac conduction abnormalities
- Hypoplastic left-heart syndrome and other left-heart abnormalities

- Sensorineural hearing loss
- Inflammatory bowel disease
- Celiac disease (increased incidence)
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Fig. 115A — Redundant nuchal skin.



Fig. 115B — Puffiness of the hands.



Fig. 115C — Puffiness of the feet.

Diagnosis

- Turner syndrome is occasionally diagnosed incidentally during prenatal testing.
- More commonly, it is suspected based upon characteristic clinical features.
- The diagnosis is confirmed by karyotype analysis.
- Prompt diagnosis is important to permit management of comorbidities, including effective treatment of short stature.
- **Indications for testing** — A karyotype analysis for Turner syndrome should be performed in any female with characteristic features, which vary by age group.

Prenatal

- Cases of Turner syndrome are sometimes discovered incidentally during chorionic-villus sampling or amniocentesis that was performed for unrelated reasons, such as advanced maternal age.

Newborn Period

- Turner syndrome may be apparent at birth, presenting with congenital lymphedema of the hands and feet, webbed neck, nail dysplasia, narrow and high-arched palate, and short fourth metacarpal.

Infants and Children

- In infants and children, Turner syndrome should be suspected in any female with unexplained growth failure, defined as a growth velocity less than the 10th percentile for age or stature that is substantially less than predicted from parental heights.

Adolescence

- Turner syndrome should be suspected in adolescent girls who fail to start or complete breast development, or those with secondary amenorrhea, especially if short stature and/or other features suspicious of Turner syndrome are present.

Diagnostic Test

- **Standard karyotype analysis** — sufficient to establish the diagnosis in most cases. However, in certain cases, the karyotype should be repeated to confirm the initial result: if the initial karyotype is normal in a patient with a strong clinical suspicion of Turner syndrome, a second karyotype should be performed using a different tissue such as the skin (fibroblasts), bladder epithelial cells in a urine sample, or buccal mucosa cells.
- Real-time PCR
- High-throughput pyrosequencing
- Whole-exome sequencing

Additional Testing

- Y chromosome mosaicism

Once the diagnosis of Turner syndrome is established by karyotype analysis, management includes evaluation and monitoring for associated abnormalities, including:

- Cardiac pathology,
- Thyroid disease,
- Hearing and eye abnormalities,
- and learning disabilities.

Age 4 Years and Older

- Serum thyrotropin (thyroid-stimulating hormone [TSH]), to screen for chronic autoimmune thyroiditis, and
- Tissue transglutaminase (tTG) with total immunoglobulin A (IgA), to screen for celiac disease.

Age 10 Years and Older

Because of the risks of diabetes mellitus, fatty liver disease, and kidney dysfunction:

- Fasting blood glucose, glycated hemoglobin (A1C),
- Alanine aminotransferase (ALT) and aspartate aminotransferase (AST),
- Serum creatinine, and urinalysis.

Key Screening Steps at Diagnosis

- Renal ultrasonography.
- Comprehensive cardiovascular evaluation by a cardiology specialist, consisting of echocardiography in infants and children and magnetic resonance imaging (MRI) in older girls and women.

Principle of Management

- Management of Turner syndrome requires an organized approach to ongoing evaluation and monitoring for associated abnormalities and prevention of common disorder.

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